THREE-ZONE OT LAB PROOF OF CONCEPT (POC)

TECHNICAL DESIGN BLUEPRINT & PROVISIONING ROADMAP

This blueprint provides the complete, end-to-end design for a Three-Zone Operational Technology (OT) Lab Proof of Concept (POC). It combines network segmentation, asset configuration, a 10-point threat matrix, and a physical process simulation engine.

1. ARCHITECTURE LAYOUT & NETWORK ALLOCATIONS

To establish defense-in-depth boundaries, the laboratory is segmented into three distinct networks managed by a centralized pfSense security gateway acting as the core conduit controller.

Zone 1: IT / Management Segment
IP: 192.168.10.5 (Ubuntu Engineering Station Workstation)

pfSense Firewall Security Gateway Layer
Interfaces: .10.1 (IT) | .20.1 (OT) | .30.1 (SIM)

Zone 2: OT Control Segment
IP: 10.0.20.10 (OpenPLC Runtime Server Engine)

Zone 3: Process Simulation Segment
IP: 172.16.30.20 (Factory I/O Physical Simulation Engine)

Detailed Network Allocations

- **Zone 1: IT / DMZ Segment (192.168.10.0/24)**
  Purpose: Engineering tasks, HMI monitoring, analytics, and security centralized logging loops.
  Primary Host: Engineering Workstation VM (192.168.10.5).
- **Zone 2: OT Control Segment (10.0.20.0/24)**
  Purpose: Controller execution layer and local automation loops. No direct external internet access is permitted.
  Primary Host: OpenPLC Runtime Engine VM (10.0.20.10).
- **Zone 3: Process Simulation Segment (172.16.30.0/24)**
  Purpose: Real-time physical industrial process rendering. Bypasses hardware costs via digital twinning.
  Primary Host: Factory I/O Windows Host / VM (172.16.30.20).

2. STEP-BY-STEP LAB PROVISIONING GUIDE

Step 1: Virtual Networking Creation

1. Open your target hypervisor platform (e.g., VMware Workstation, Proxmox VE, or VirtualBox).
2. Instantiate three isolated virtual host-only networks or custom local LAN segments:
    - `LAN_IT_ZONE` (Host-Only / Custom)
    - `LAN_OT_ZONE` (Host-Only / Custom)
    - `LAN_SIM_ZONE` (Host-Only / Custom)

Step 2: Firewall Setup & Interfaces (pfSense)

1. Provision a new virtual appliance with 1 vCPU, 1GB RAM, and 3 Network Adapters enabled.
2. Bind the network adapters sequentially inside your hypervisor settings:
    - Adapter 1 → `LAN_IT_ZONE`
    - Adapter 2 → `LAN_OT_ZONE`
    - Adapter 3 → `LAN_SIM_ZONE`
3. Boot the pfSense appliance. Using the command console interface on the local system, configure the static IPv4 parameters:
    - Interface 1 (WAN / IT): **192.168.10.1/24**
    - Interface 2 (LAN / OT): **10.0.20.1/24**
    - Interface 3 (OPT1 / SIM): **172.16.30.1/24** (Enable this interface via the web GUI and rename it to `SIM_ZONE`).

Step 3: Zone 2 Control Node Provisioning (OpenPLC)

1. Deploy an Ubuntu Server 22.04 LTS VM bound directly to the `LAN_OT_ZONE` interface segment.
2. Map static network parameters within the Netplan configuration layout file:

```
network:
  ethernets:
    eth0:
      addresses: [10.0.20.10/24]
      gateway4: 10.0.20.1
      nameservers:
        addresses: [10.0.20.1]
  version: 2
```

3. Run the execution sequence: `sudo netplan apply`.
4. Download and install the OpenPLC Runtime framework:

```
git clone https://github.com/thiagoralves/OpenPLC_v3.git
cd OpenPLC_v3
./install.sh linux
```

5. Reboot the server, access the administrative console via a web browser at `http://10.0.20.10:8080`, and verify that Modbus port 502 is listening.

Step 4: Zone 3 Simulation Node Provisioning (Factory I/O)

1. Boot a Windows 10/11 system bound strictly to the `LAN_SIM_ZONE` segment.
2. Configure IPv4 properties on the local network adapter manually:
   IP Address: 172.16.30.20 | **Subnet Mask:** 255.255.255.0 | **Gateway:** 172.16.30.1
3. Download and open Factory I/O. Navigate to **File > Drivers**, then select **Modbus TCP Server**.
4. Click **Configuration**. Set Network Interface to `172.16.30.20`, Port to `502`, and allocate 16 digital inputs and outputs for logic sequencing.

Step 5: Zone 1 Management Node Provisioning (Engineering Workstation)

1. Deploy an Ubuntu Desktop VM bound to the `LAN_IT_ZONE` segment.
2. Assign the static parameters: IP `192.168.10.5`, netmask `255.255.255.0`, gateway `192.168.10.1`.
3. Install simulation management and diagnostic utilities:

```
sudo apt update && sudo apt install snapd -y
sudo snap install qmodmaster
```

3. CONDUIT & FIREWALL ACCESS CONTROL LISTS (ACLS)

Implement these explicit security filtering rules within the pfSense interface (**Firewall > Rules**) to establish a strict, zero-trust structural architecture between security conduits:

SOURCE INTERFACE	SOURCE IP	DEST INTERFACE	DEST IP	PROTOCOL / PORT	ACTION	PURPOSE
IT_ZONE	192.168.10.5	OT_ZONE	10.0.20.10	TCP / 502	**PASS**	Allows SCADA / Modbus polling from Engineering Workstation to PLC.
IT_ZONE	192.168.10.5	OT_ZONE	10.0.20.10	TCP / 8080	**PASS**	Allows PLC web console administrative dashboard access.
OT_ZONE	10.0.20.10	SIM_ZONE	172.16.30.20	TCP / 502	**PASS**	Allows PLC to command/read physical values on the Factory I/O engine.
ANY	ANY	ANY	ANY	ANY	**DENY**	Implicitly drops all cross-zone traffic traversing conduits.